

Library Management System Using QR(LMS)

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Abstract—Library management systems (LMS) are designed to automate the various processes involved in managing a library, such as tracking books, issuing them to members, and maintaining records of borrowing activities. This paper explores the design, implementation, and evaluation of a Library Management System (LMS) that aims to optimize library operations. By using modern technologies such as relational databases and web-based interfaces, the system offers features like book cataloging, member management, and transaction processing. This paper also discusses the challenges faced in implementing such systems and presents recommendations for improvement.

I. INTRODUCTION

In this digital world in advances in communication technology data and information is no longer tied to physical location. Current generation students often adapt at multiprocessing and make use of internet resources rather than offline library resources. This can be attributed to an ease of finding information in one click rather than spending time in searching books. The primary issue for any library system is organization and record keeping. Due to large no. of users and items there are chances of misplacement of books due to frequent movement of books within or outside of library. The struggle for librarians and users to find the books. A library management system is software that is designed to manage all the functions of a library. It helps librarians to maintain the database of new books and the books that are borrowed by members along with their due dates. The best way to maintain, organize, and handle countless books systematically is to implement a library management system software.

To maintain integrity, it's essential that all QR codes follow a consistent format for each book or resource. This ensures that QR codes can be easily scanned and interpreted by the system. The library management system should automatically update whenever a QR code is scanned, ensuring that the latest information is available.

Identify applicable funding agency here. If none, delete this.

The objective of the study (improving library management through a barcode system). A brief description of how barcode scanning can automate and streamline tasks such as inventory tracking, check-in/check-out, and resource management. The user should be able to search any books stored in the database. Users can search the year, title, publisher, author.. Librarians can manage books and transactions with minimal effort, while members can quickly search for books and view their borrowing history. Library Management System using QR code. Scanner contains important information such as the book's title, author, or the user's ID and borrowing history. By scanning the QR code using a smartphone or scanner, library staff and users can quickly access and update records. Scanner allows for faster processing of book issues and returns, reducing wait times for user.

This system completely automates all your library's activities. The best way to maintain, organize, and handle countless books systematically is to implement a library management system software. You can find books in an instant, issue/reissue books quickly, and manage all the data efficiently and orderly using this system. The purpose of a library management system is to provide instant and accurate data regarding any type of book, thereby saving a lot of time and effort. We must use the Library Management System in order to have secured storage of book details contained within the Library. This feature is generally enabled in order to protect the data, which is highly confidential.

II. LITERATURE REVIEW

2.1 Majid Bayani, Alberto Segura, Marjorie Alvarado, Mayra Loaiza "IoT-Based Library Automation and Monitoring system: Developing an Implementation framework of Implementation" Volume 8, number 1 Technical Article 3 Jan-Jun 2018 e-Ciencias de la Información.

Concluded: Using the mobile applications is a trend. People are spending high percentages of their time using their smart

phones and mobile apps, then applying a library mobile application can increase the users' library usage rate. It is very common to use the smart phones to connect to the Internet, to buy a product, to pay a service, to watch a video and to get access to numerous online services. One of the advantages related to the library service management which the IoT can offer is that

once the users download and install the library application, they can have access to all online services prepared by the system. It can play a key role in the human's global data access and knowledge propagation in a fast, more efficient and smart manner.

2.2 Suhas Holla, Mahima M Katti, "Android based mobile application development and its security". International Journal of Computer Trends and Technology- volume3 Issue3-2012

Concluded: In this paper we concluded that the mobile application is very important and simplest use and convenient to user because all applications are done by Java Script language. A small application that runs on a smartphone and a tablet is called an app. The speed of applications or page loading mustn't keep users waiting. However, issues such as this, are usually checked by quality assurance, which is a part of a smart mobile application or software development. Security is the main concern of any software application. Security related problem faced by the many of the mobile user's this is vital to many apps. However, issues such as this, are usually checked by quality assurance, which is a part of a smart mobile application or software development. Security is the main concern of any software application. Security related problem faced by the many of the mobile user's this is vital to many apps. It is one of the first topics of discussion between you and software developers. No leaks of the users' private information are allowed. Support and updates

are one of the primary issues. To build a long- standing app, you need to consider support and

updates. Maintain the server. Ensure that your content is made up of up-to-date, relevant information. Analytics is essential feature of a software application. As a mobile app developer, one key component is to incorporate analytics into your mobile app. [2]

2.3 U. Narmadhaa, P. Pavithra, M. Tharuneswari, S. Sowmiya, Nagarajan. "Enhanced QR-code based application for library management system using android". International Journal on Applications in Information and Communication Engineering Volume 3: Issue 1: February 2017, pp 46-49.

Concluded: QR code (abbreviated from Quick Response Code) is the trademark for a type of matrix barcode (or two-dimensional barcode) first designed for the automotive industry in Japan. A QR code consists of black squares arranged in a square grid on a white background, which can be read by an imaging device such as a camera, and processed using Reed–Solomon error correction until the image can be appropriately interpreted. The required data is then extracted from patterns that are present in both horizontal and vertical components of the image. QR codes features include high

reading speed, data storage and transfer, 360degree readability, resistance to contamination and defects and adding up to 16 symbols to the structure. [3]

textbf2.4 Lambodara Parabhoi, Nivedita Bhattacharjya, Rupashree Dhar "Use of QR Code in Library" researchgate.net/publication/318259063 January 2017 Concluded: Libraries are now being very challenged by the development of various technologies. In this present era, the new technology like QR code demands the changes of information handling in the library. The user would have easy access to most current and necessary information related to the library by using QR code. So to make the effective use of QR code among the user community, library professionals must organize user awareness program, orientation program, etc.

2.5 Mr. Nagesh L. Londhe, Dr. Suresh K. Patil "Open source library management systems: a survey and present developmental status" International Journal of Library and Information Science (IJLIS) Volume 4, Issue 1, January - April (2015), pp. 38-54

Concluded: This study covers thirty-one open source library management systems projects, which are developed during 1999 to 2014 and tries to present the state of development. Researcher has also attempted to present

historical overview of development and update picture of library management systems. This is one of the comprehensive up-to-date survey. This study reveals that only fifteen LMS projects shown recent activity in community and their releases. Fourteen projects do not show any activity for than four years since their last activity (release and community) upto the date of data collection (Dec. 2014). Two projects developmental status could not be determined due to unavailability of data. This study reveals that maximum number of active projects is having an institutional support. This study also reveals there are considerable number of weekly downloads for old releases of some inactive state projects.

Madhusudan (2022) studied the increasing usage of QR codes in marketing and other industries, emphasizing their role in simplifying information access via mobile devices. Das and Das (2021) highlighted QR codes as a revolutionary tool in libraries, aiding in book issuance, instructions, and resource management. Similarly, Bellary (2018) emphasized QR codes as an effective means of accessing information without memorizing URLs, thus streamlining library interactions.

Dadhich, Hiran, Rao, Sharma, et al. (2022) explored the transformation of traditional library management systems into smart libraries using QR codes. Their study found that QR-based systems significantly reduce librarians' workload, improve book tracking, and provide real-time data management. Another study by Dadhich, Poddar, et al. (2022) examined the rapid adoption of QR technology in Indian libraries, attributing its increasing acceptance to the widespread use of smartphones and mobile-based services.

Applications of QR Code Technology

Several studies have examined the practical applications of QR codes in libraries. Gaurav Kumar Singh (2022) explored QR code integration into library services, discussing

its advantages and limitations. The study identified QR codes as a "treasure house of knowledge" that simplifies access to digital and physical library resources. Similarly, Butler et al. (2021) highlighted how QR codes provide instant access to high-quality academic resources with minimal effort. Further, Dadhich, Doshi, et al. (2022) and Panigrahi Sethy (2022) emphasized the role of QR codes in change management within library and information sciences. Their research advocated for the adoption of new technologies to improve efficiency and cater to users' needs. The technical aspects of QR codes have been widely discussed. Studies suggest that QR codes provide high-capacity data encoding, small printout sizes, and resistance to damage (Dadhich, Manish, Shalendra Singh Rao, Renu Sharma, 2021). These features ensure that QR codes remain functional even when partially damaged, making them a reliable tool for library management. Moreover, researchers have highlighted the ease of implementation and user-friendliness of QR codes. Open-source QR code generators and pre-installed scanners in smartphones have facilitated the widespread adoption of this technology (Bellary, 2018; Purohit et al., 2022). QR codes have also been recognized for their role in document delivery services, increasing information literacy, and linking users to essential library resources (e.g., Web-OPAC, Institutional Digital Repository, and e-books).

Conclusion The existing literature demonstrates that QR codes are a powerful tool for modernizing library management and improving user experiences. Research findings collectively indicate that QR codes enhance accessibility, reduce manual effort, and support mobile-based services. Their adoption in libraries is expected to grow as technological advancements continue to improve user engagement and operational efficiency.

A. CHALLENGES AND SOLUTIONS

Data Integrity: Ensuring the integrity of book and transaction data was challenging. This was addressed by implementing strict database constraints and triggers to prevent data anomalies.

User Authentication: Ensuring secure and efficient authentication for both librarians and members was critical. This was solved using secure password storage methods and session management techniques.

System Scalability: As the library grows, the system needs to handle more books and members. The database was optimized for scalability by using indexing and efficient query processing.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem statement

Library management system provides a simple GUI (Graphical User Interface) for the Library staff to manage the functions of the library effectively. The Library Management system is designed and developed for the receipt and Issuance of books in the library. This case study on the library management

system gives us the complete information about the library and the daily transactions done in a Library. We need to maintain the record of news the details of books available in the library which mainly focuses on basic operations in a library like adding new member, new books, and up new information, searching books and members and facility to borrow and return books. Though some libraries do have computerised systems that provide basic features such as adding books, user etc and also that of issuing the books, but what they lack is the user involvement and interactivity. When computerized system is not implemented file is always lost because of human environment. Sometimes due to some human error there may be a loss of records. File damaged When a computerized system is not there file is always lost due to some accident like spilling of water by some member on file accidentally. Besides some natural disaster like floods or fires may also damage the files. When there is no computerized system there is always a difficulty in searching of records if the records are large in number.

B. Objective

- Automate library operations using barcode scanning.
- Improve book search and retrieval functionality
- Enhance circulation management and fine tracking.
- Provide secure and easy access to library services

IV. METHODOLOGY

A Library Management System by combining the frontend, a backend service (Supabase), and an external book database (Google Books API). I created the user interface using React and TypeScript, making sure it's clean and easy to use with Tailwind CSS. I set up user login and authentication using Supabase, so users can securely sign in and manage their borrowed books. I also connected the system to the Google Books API, which allows users to browse and search for books in real time. To keep everything organized, I used React hooks to manage user authentication and app state. I also built a borrowing system that updates the database, so users can track the books they've borrowed. Overall, I made sure the system is well-structured, user-friendly, and works smoothly by handling API requests properly and ensuring fast responses.

Programming Languages : TypeScript (for the frontend, with React)

Database : PostgreSQL (via Supabase)

Front-end : React (with TypeScript Vite) , Tailwind CSS

Backend : Supabase (Backend-as-a-Service)

External API : Google Books API – Fetches book data dynamically

V. FIGURE

1. **User Interface** This is the front-end layer where users interact with the system. It provides an interface for logging in, searching for books, borrowing/returning books, and accessing other system features. It can be a web application,

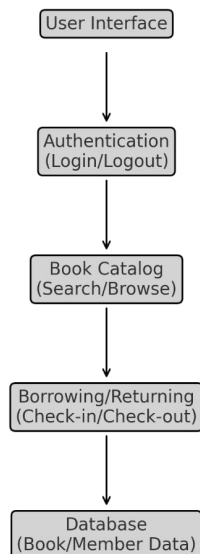
mobile app, or desktop application. Users such as members and administrators interact through this layer.

2. Authentication (Login/Logout) Ensures that only authorized users can access the system. Users need to log in with valid credentials (username and password). Admins may have special privileges to manage book catalogs or handle member data. After successful authentication, users gain access to the relevant features.

3. Book Catalog (Search/Browse) This module allows users to search, browse, and view available books in the library. Users can search using parameters such as title, author, genre, or ISBN. It may display information like book availability, location, and reservation status. Advanced catalog features might include filters or sorting options.

4. Borrowing/Returning (Check-in/Check-out)

Handles the process of borrowing and returning books. Users can check out available books, and their data is recorded. When books are returned, the system updates the status to make the books available for other members. This module may also handle overdue books, fines, and notifications.



[h]

Flow Chart of LMS

At the center, a welcoming message, "Welcome to Your Digital Library," invites users to explore the platform. Below the main message, users are encouraged to either "Create Account" or "Sign In," highlighting the system's focus on personalization and user management.

The interface also showcases three core features of the platform:

Vast Collection – Emphasizes the availability of a large

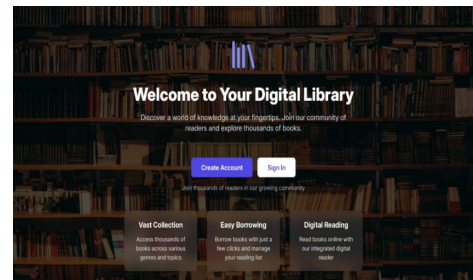


Fig. 1. LMS User Interface Homepage

number of books across different genres and topics.

Easy Borrowing – Allows users to borrow books digitally with a few clicks, streamlining the lending process.

Digital Reading – Provides integrated tools for reading books directly within the platform, eliminating the need for external applications

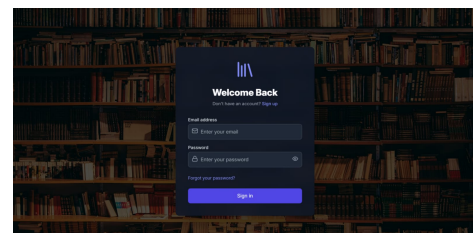


Fig. 2. User Interface For LMS

Interface Design of a Library Management System (Login Page)

Email Field: Requires users to input their registered email.

Password Field: Password input with an option to toggle visibility.

Forgot Password: A link that allows users to recover access.

Sign-In Button: Action button to authenticate and access the system.

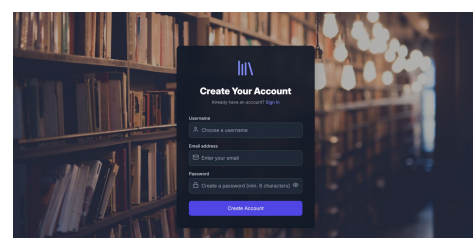


Fig. 3. Create Your Account For LMS

Username – Input field to choose a unique username.

Email address – Field to enter a valid email

. Password – Field to set a password (minimum of 6 characters), with an eye icon to toggle password visibility.

Button: "Create Account" – Purple button to submit the form and create a new account.

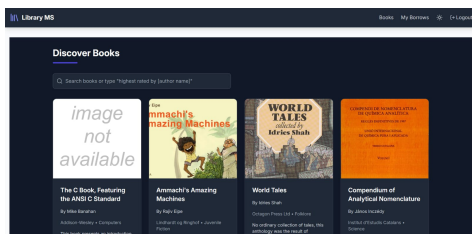


Fig. 4. User Interface of the LMS

The image above shows the Discover Books section of a Library Management System web application. This interface enables users to search for and browse books available in the library collection. It features a responsive UI with essential navigation options, such as “Books”, “My Borrows”, and “Logout”, located at the top right corner.

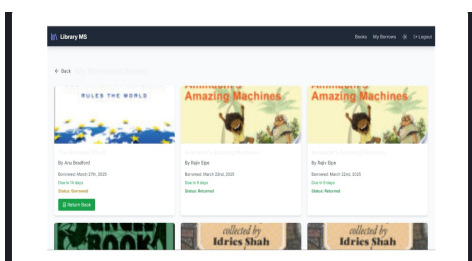


Fig. 5. Borrowed Books Dashboard Interface LMS

The image showcases a key feature of the digital library system: the “My Borrowed Books” dashboard. This section enables users to view and manage their currently borrowed digital books efficiently. The interface is user-friendly, visually clean, and organized into individual cards for each borrowed book.

Each book card displays the following details:

Book Title and Author Borrowed Date

Due Date

Borrowing Status (e.g., “Borrowed”, “Returned”)

Return Option – For books still borrowed, a green “Return Book” button is provided, allowing easy manual return.

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A. CONCLUSION

This paper presented the design and implementation of a Library Management System that automates key library operations. The system significantly reduces the administrative burden on library staff and provides a more efficient service to members. While the system has been successful in addressing the key requirements, future work will focus on integrating

features like book reservation, remote access, and mobile support.

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The SLMS application and QR code implementation has been built and tested: The login, profile data, library management and QR code have been successfully implemented. The homepage of the application also gives the user options to choose between the librarian or the student edition. Each student user has a personal page which contains their information and the history of books they issued with the library. This allows the user to choose between past interests and track movement of the books they have issued. Each staff user gets access to the list of books in and out of the library. Whenever a book is issued by a user and taken out of the library, it is added to the list of books away. The system was tested with a sample data set of 20 books. The test user was successfully able to issue and return a book using a smartphone and search for books as per convenience.

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